22.9.26 Experiment - 8.

Aim: To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP sca and TCP/UDP ping scan - and to understand wh protocol at which layer of TCP/IP model, and when one applies.

Introduction

Nmap (Network mapper) is free-open source network scanning tool created by Gordon Lyon & just realsed in 1997. It is the Industry standard tool for mapping networks, discovery live hosts.

(1) ARP (Link Layer) - for hosts on the same local subnet

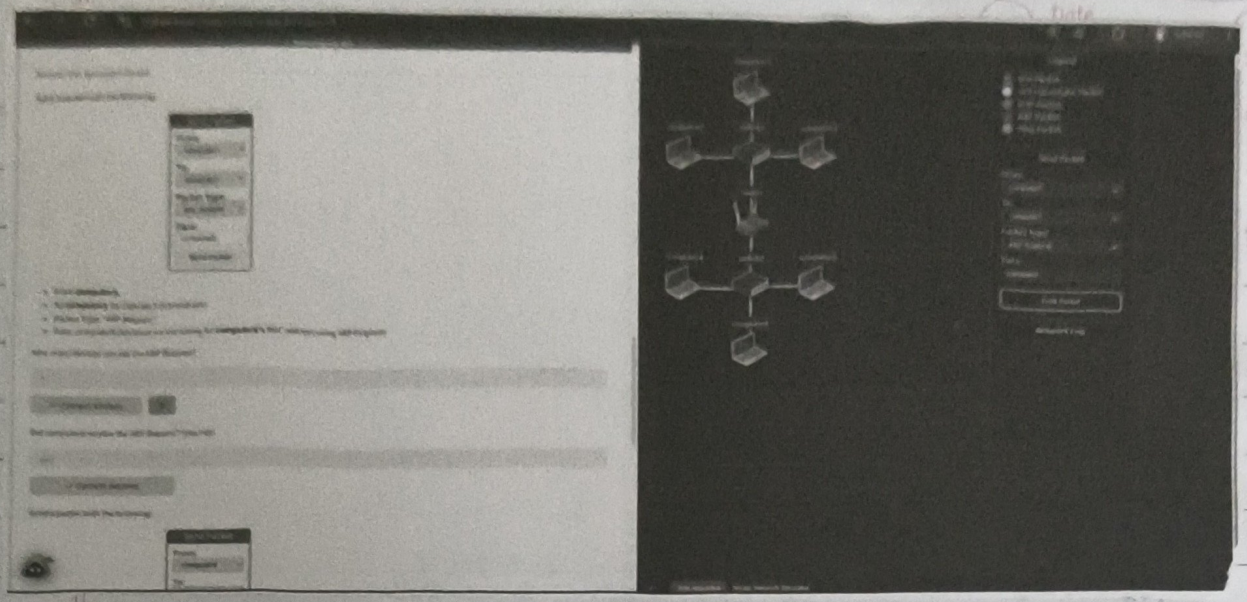
(2) ICMP (network layer) - echo timestamp address mask requests

(3) TCP (transport layer) - SYN, ACK ping probes

(4) UDP (transport layer) relies on ICMP reply

(1) ARP host discovery.

On a local element or wifi communication reciever destination MAC address no learn it a host broadcast an ARP resources any host holding that IP replies with its MAC address. A reply proces that host is up. ARP frames are not routed. By default a user scanning a local subnet automatically uses ARP



(2) ICMP Host Discovery:

ICMP offers several resources that can reveal a live host. Windows systems, many firewall block ICMP by default. So Nmap also supports timestamp address - mask resources as fallback.

7.1 ICMP Echo (-PE)

Nmap -PE ... 10.200.6.0/24 Nmap scan report for 20.200.6.1 Host is up (0.06s)

Nmap scan report for 20.200.6.1

Nmap scan report for 16-200-8.6.200 Host is up (0.70s latency)

Nmap scan timestamp (-PP)

7.3 ICMP Address Mask (-PM)

Most commonly filled.

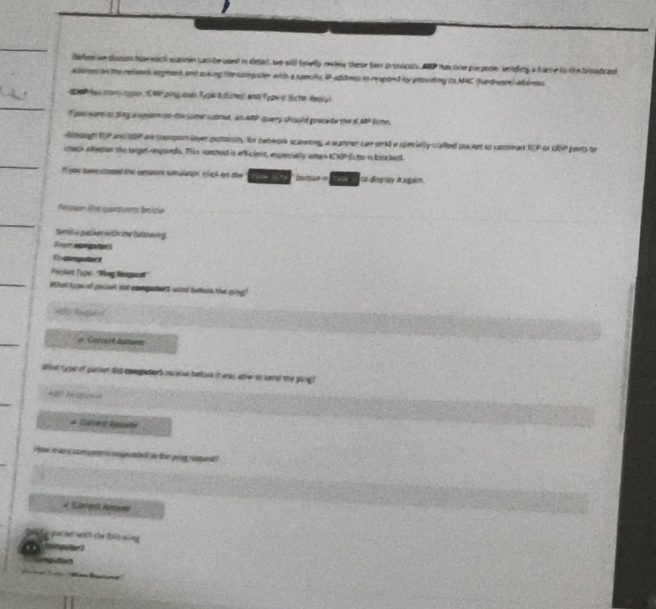
$nmap -pm -sn 10.200.6.0/24

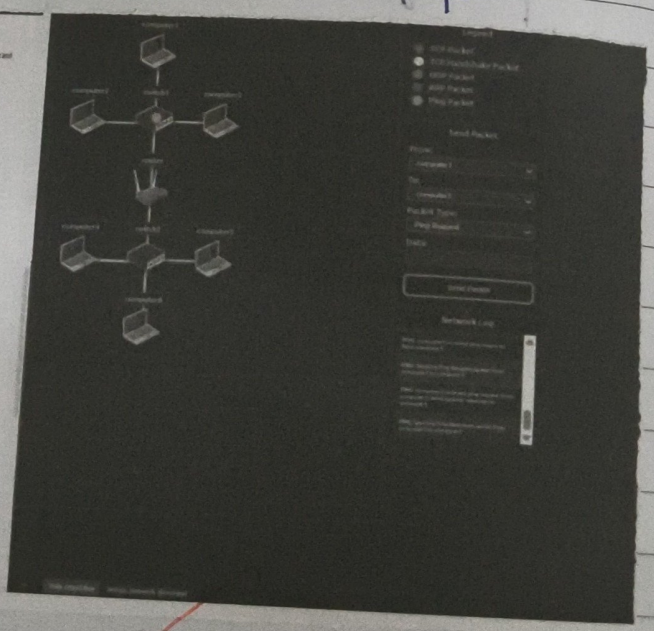
Nmap dr: 250 Ip address scanned in 52.14 seconds.

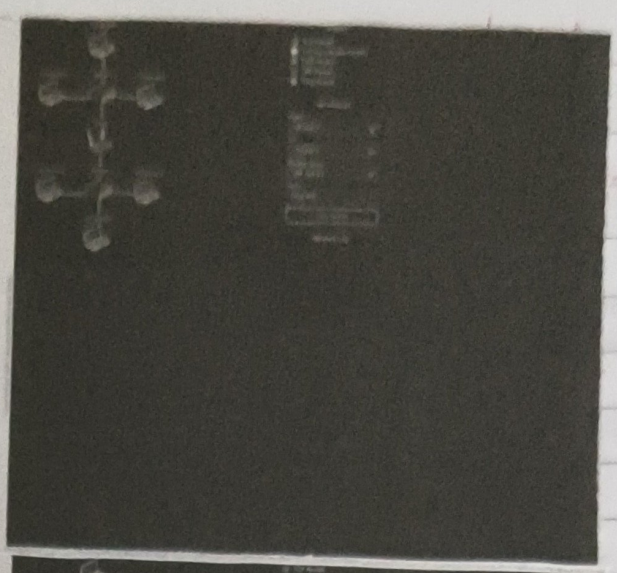
TCP, UDP ping Discovery.

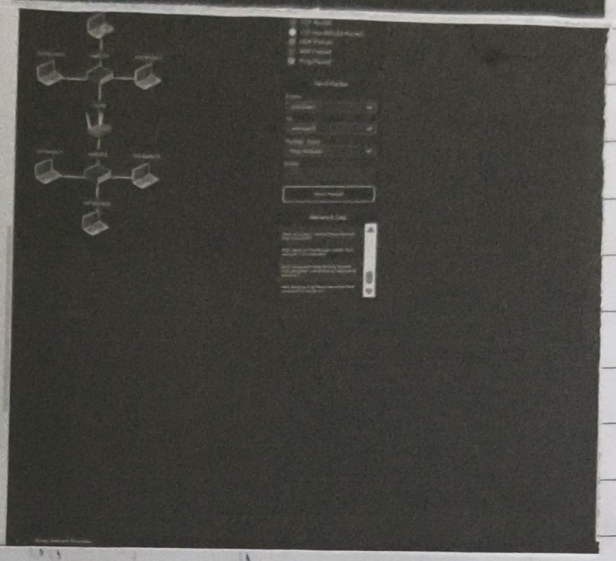
When ICMP is filtered. Nmap can send crafted TCP or UDP packets, infer that a host is alive purely from the fact spreads at all.

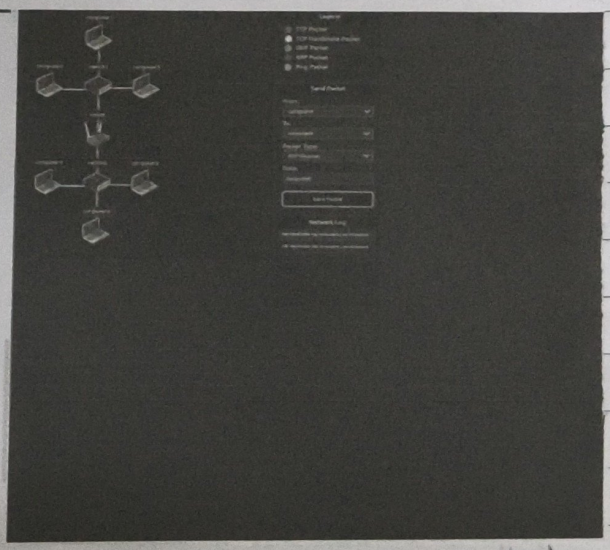
TCP SYN ping (-PS)
Sends a TCP packet with SYN flag To
port - An open port SYN ACK a closed
port output RST Specify port as
PS80, PS21, or PS80
TCP ACK ping (-PA)
Sends a TCP packet with Ack flag set
Because Ack does not belong no any
this a not sudo user.
$sudo nmap -PA -sn 10.200.6.0/24
Nmap Scan report for 10.200.6.50
Host is up (0.70s latency)
Nmap Scan report for 10.200.6.250
Host is up 10.069 (latency)
$sudo nmap -PO -sn 10.200.6.0/24
Nmap Scan report for 10.200.6.250
Host is up 10.069s (latency)
Nmap done: 256 IP addres (2 hosts up)











Result:

Therefore NMAP host discovery using Tryhackme implemented successfully